FRONT LOCK INDICATOR (Continued)

Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the "FRONT LOCK" text to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The illumination intensity of the front lock indicator is one-step dimmable. When the exterior lighting is turned On, the indicator is dimmed; and, when the exterior lighting is turned Off, the indicator is illuminated at full intensity. The front lock indicator is serviced as a unit with the instrument cluster.

OPERATION

The front lock indicator gives an indication to the vehicle operator of the status of the locker mechanism in the front axle of vehicles equipped with the optional off-road package. The front lock indicator can also give an indication when certain faults are detected in the axle locker system. This indicator is controlled by a transistor on the instrument cluster electronic circuit board based upon cluster programming, and hard wired inputs to the cluster from the key-in ignition switch, the axle lock switch, and the front locker indicator switch. The front lock indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will allow this indicator to operate whenever the instrument cluster receives a battery current input on the fused B(+) circuit. Therefore, the LED can be illuminated regardless of the ignition switch position. The cluster can illuminate the LED solid, or flash it on and off at two different rates, slow or fast. The slow rate is about one flash per second, while the fast rate is about two flashes per second. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the front lock indicator for the following reasons:

- **Front Axle Lock Request Input** - Each time the cluster detects ground on the front axle lock request circuit from the axle lock switch (front axle lock request circuit ground = front axle lock request active) while the key is in the ignition lock cylinder, the front lock indicator flashes on and off at a slow rate. The indicator continues to flash until the front axle lock request input to the cluster is an open circuit (front axle lock request circuit open = front axle lock request inactive), or until the cluster detects a change in the inputs from the front locker indicator switch or the key-in ignition switch, whichever occurs first.
- **Front Locker Indicator Switch Input** - Each time the cluster detects ground on the front locker indicator switch sense circuit (front locker indicator switch closed = front axle locker engaged) while the front axle lock request is active and the key is in the ignition lock cylinder, the front lock indicator is illuminated solid. The indicator remains illuminated solid until the front locker indicator switch input to the cluster is an open circuit (front locker indicator switch open = front axle locker disengaged), or until the cluster detects a change in the inputs from the axle lock switch or the key-in ignition switch, whichever occurs first.
- **Front Lock Request/Feedback Mismatch** - Each time the cluster detects that the inputs from the axle lock switch (request) on the front axle lock request circuit and from the front locker indicator switch (feedback) on the front locker indicator switch sense circuit are mismatched, the front lock indicator flashes on and off at a slow rate. This condition occurs when the lock request is active but the axle locker remains disengaged, or when the lock request is inactive but the axle locker remains engaged. The indicator continues to flash until the request and the feedback are matching.
- **Key-In Ignition Switch Input** - Each time the cluster detects that the key has been removed from the ignition lock cylinder (key-in ignition switch open = key is not in ignition switch) while the front axle lock request is still active, the front lock indicator flashes at a fast rate and three chime tones are sounded. The indicator will continue to flash at this rate until the key is replaced in the ignition lock cylinder (key-in ignition switch closed = key is in ignition switch) while the front axle lock request is still active, or until the front axle lock request input is deactivated, whichever occurs first.

The axle lock switch in the accessory switch bezel on the instrument panel provides a hard wired ground input to the instrument cluster circuitry through the front axle lock request circuit whenever the switch is enabled and the rear and front axle lock position of the switch is selected by the vehicle operator. The front locker indicator switch on the front axle housing provides a hard wired ground input to the instrument cluster whenever the front axle locker mechanism is engaged. The key-in ignition switch integral to the ignition switch provides a hard wired ground input to the instrument cluster whenever a key is present in the ignition lock cylinder. Each of these inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. For further diagnosis of the instrument cluster circuitry that controls the front lock indicator LED, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING).